

SAMPLING and QUNTIZATION

Introduction

- To create a digital image, we need to convert the continuous sensed data into digital form.

This process includes 2 processes:

- **Sampling:** Digitizing the co-ordinate value is called sampling.
- **Quantization:** Digitizing the amplitude value is called quantization.

Image Sampling

- The process of digitizing the co-ordinate values is called Sampling.

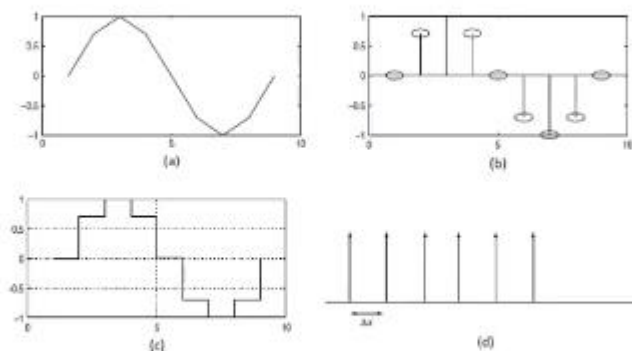


Fig. 2.11 Sampling process (a) Original signal $f(t)$ (b) Sampled image $f(n) = f(t) * r(t)$ (c) Reconstructed image (d) Train of the impulse function $r(t)$

Shannon–Nyquist theorem

The sampling frequency should be greater than or equal to $2 \times f_{\max}$, where $f_{\max}$ is the highest frequency present in the image.

- A continuous image $f(x, y)$ is normally approximated by equally spaced samples arranged in the form of a $N \times M$ array where each element of the array is a discrete quantity.
- The sampling rate of digitizer determines the spatial resolution of digitized image.
- Finer the sampling (i.e. increasing M and N), the better the approximation of continuous image function $f(x, y)$.

$$f(x,y) = \begin{bmatrix} f(0,0) & \dots & f(0,M-1) \\ \vdots & \ddots & \vdots \\ f(N-1,0) & \dots & f(N-1,M-1) \end{bmatrix}$$

Sampling

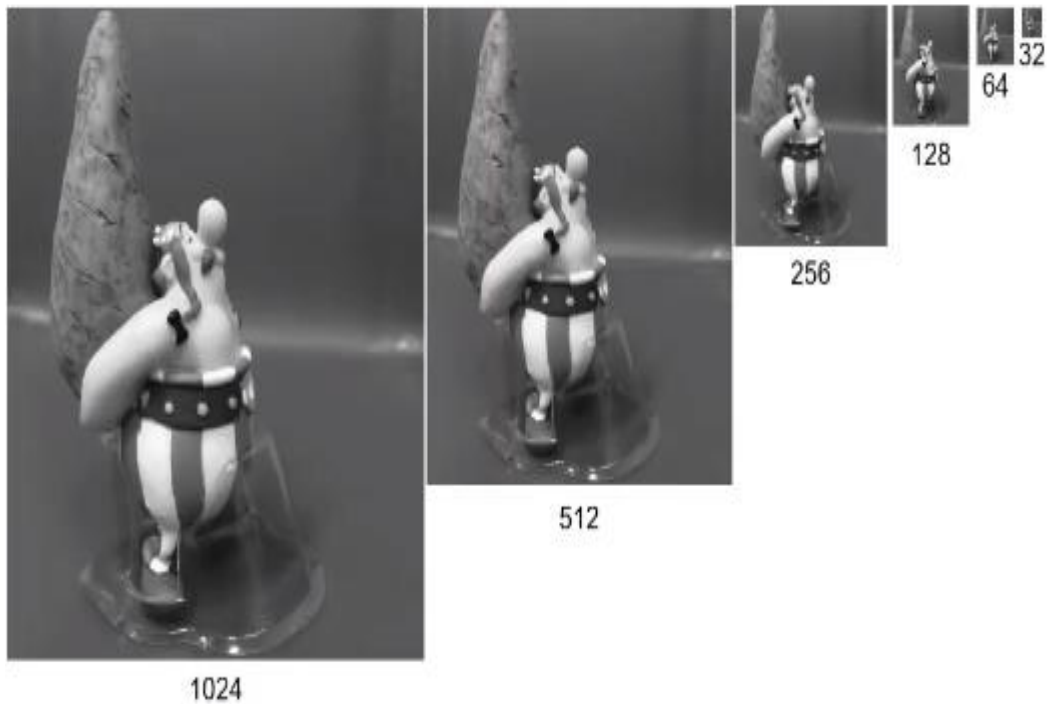
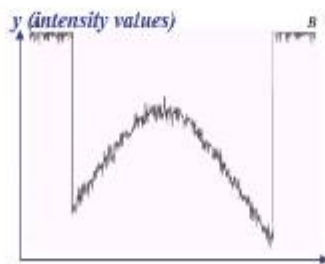
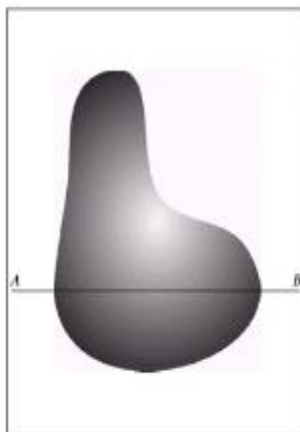
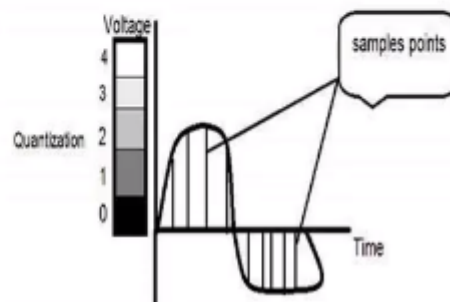


Image Quantization

- The process of digitizing the amplitude values is called Quantization.
- No of quantization levels should be high enough for human perception of the fine details in the image.
- Most digital IP devices uses quantization into k equal intervals.
- If b-bits are used,

$$\text{No. of quantization levels} = k = 2^b$$

- 8 bits/pixels are used commonly.



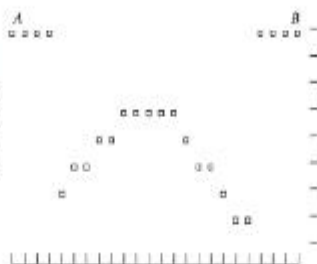
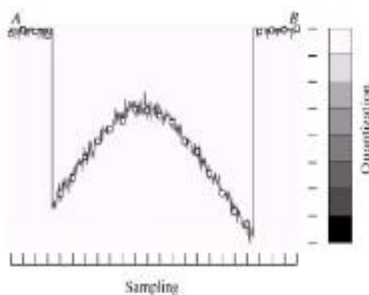
Steps for generating a digital image.

(a) Continuous image.

(b) A scaling line from A to B in the continuous image, used to illustrate the concepts of sampling and quantization.

(c) Sampling & Quantization.

(d) Digital scan line.



- Practically method of sampling is determined by the sensor arrangement used to generate the image.
- Sensors may be a single or strip of sensors as per the requirement.
- Quantization of sensor outputs complete the process of generating a digital image.
- The quality of digital image is determined to a large degree by the number of samples and discrete gray levels used in sampling and quantization.

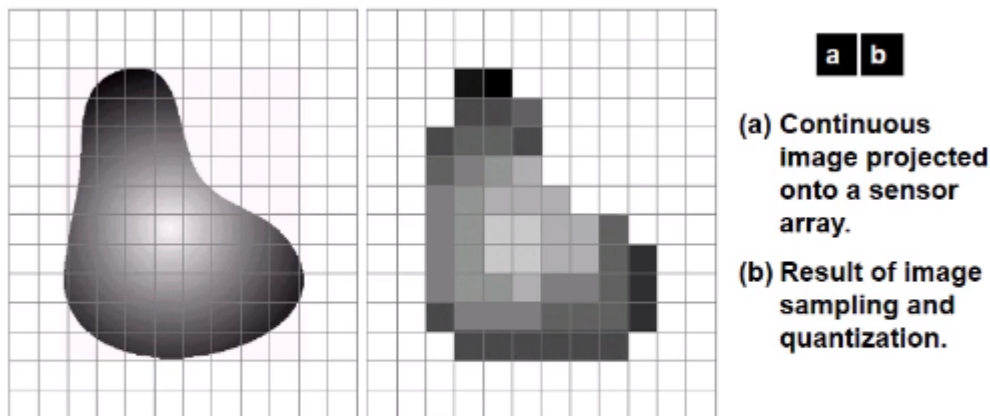
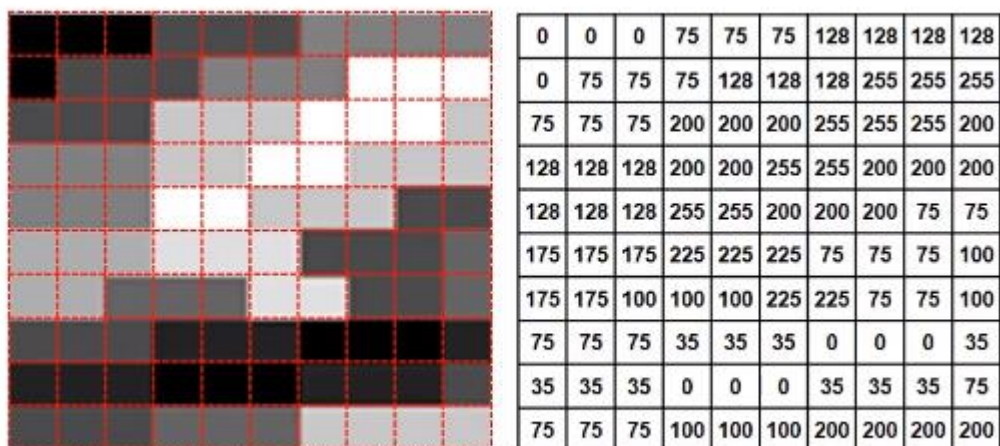
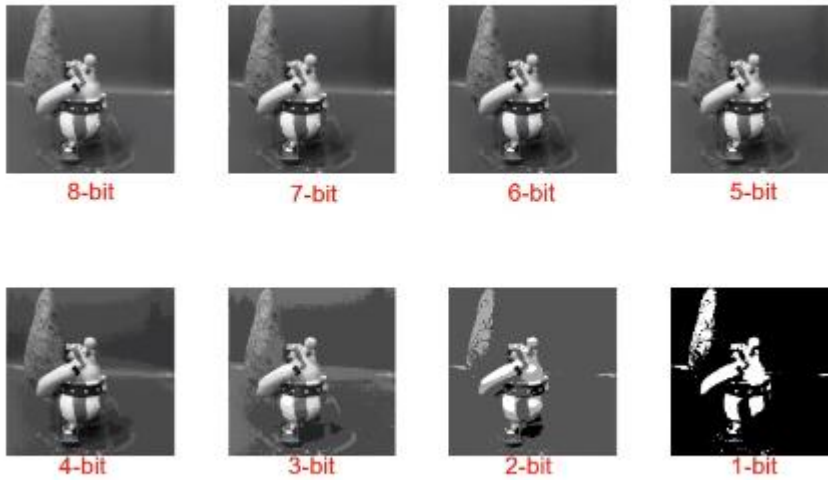


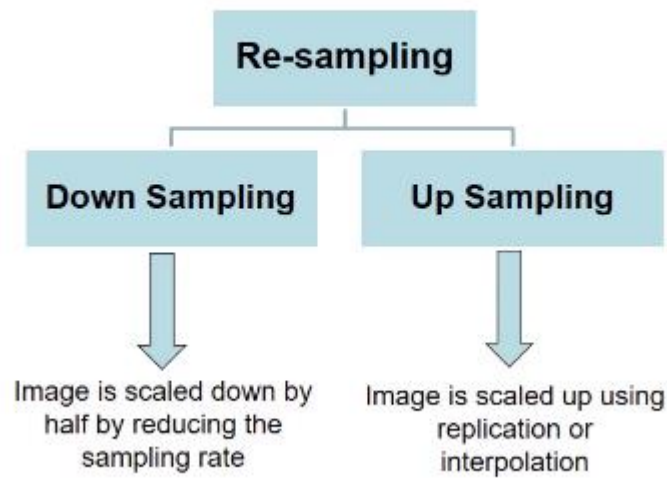
FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.



Quantization



Types of sampling



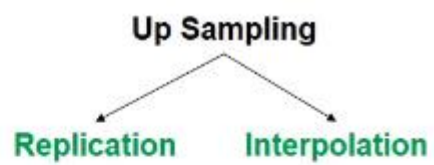
Down Sampling

F =

3	3	3	3
9	9	9	9
3	3	3	3
9	9	9	9

F =

2	1
1	3



F =

2	1
1	3

Interpolation

Basic Relationship between Pixels

Neighbors of a Pixel

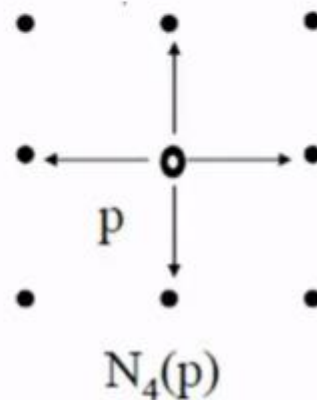
- N_4 - 4-neighbors (2 Horizontal, 2 Vertical)
- N_D - diagonal neighbors
- N_8 - 8-neighbors ($N_4 \cup N_D$)

Neighbors of a Pixel

- $N_4(P)$: 4-Neighbors of Pixel P are
 - 2 vertical neighbors
 - 2 horizontal neighbors

Coordinates of neighbors of P (x,y) are
(x+1, y), (x-1, y), (x, y+1), (x, y-1)

Distance from p
Each of the neighboring pixels in $N_4(P)$ are at a unit distance from P



Neighbors of a Pixel

▪ $N_D(P)$: Diagonal Neighbors of Pixel P

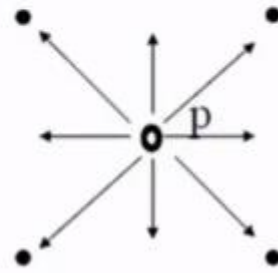
- The four diagonal neighbors of $p(x, y)$ are given by

$$(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)$$

- This set is denoted by $N_D(P)$

Distance from P:

Each of neighboring pixels are at Euclidean distance of 1.414 from P



$N_D(P)$: Diagonal Neighbors of Pixel P

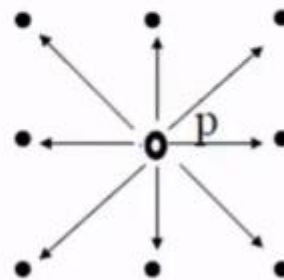
Neighbors of a Pixel

▪ $N_8(P)$: 8-Neighbors of Pixel P are

- The points $N_D(P)$ and $N_4(P)$ are together ($N_D(P) \cup N_4(P)$) known as 8-neighbors of the point P

- It is denoted by $N_8(P)$

- 2 vertical neighbors
- 2 horizontal neighbors
- 4 diagonal neighbors



$N_8(p)$

▪ Boundary Case

- Some of the points in the N_4 , N_D and N_8 may fall outside image when P lies on the border of image

Neighbors of a Pixel

- N_4 - 4-neighbors (2 Horizontal, 2 Vertical)
- N_D - diagonal neighbors
- N_8 - 8-neighbors ($N_4 \cup N_D$)

N_D	N_4	N_D
N_4	P	N_4
N_D	N_4	N_D

Adjacency

- **Property**
 - Two pixels are connected if they are neighbors, and
 - their gray levels satisfy some specified criterion of similarity
- Example: in a binary image two pixels are connected
 - if they are 4-neighbors and
 - have same value (0/1)

Adjacency

- Various types of adjacency
 - 4-adjacency
 - 8-adjacency
 - m-adjacency
- To understand adjacency let us define V
 - Let V be set of gray levels used to define adjacency

$$V = \{\text{set of gray levels}\}$$

Adjacency

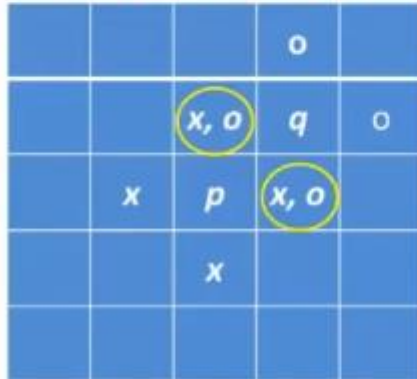
- 4-adjacency:
 - Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$
- 8-adjacency:
 - Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$
- m-adjacency (mixed adjacency):
 - Two pixels p and q with values from V are m-adjacent if,
 - q is in $N_4(p)$ **or**
 - q is in $N_D(p)$ and the set $[N_4(p) \cap N_4(q)]$ is empty (has no pixels whose values are from V)

Adjacency

▪ m-adjacency (mixed adjacency):

– Two pixels p & q with values from V are m-adjacent if

- q is in $N_+(p)$ **or**
- q is in $N_D(p)$ and the set $[N_+(p) \cap N_+(q)]$ is empty (has no pixels whose values are from V) [yellow circles should not have anything common]



x : neighbor of p : $N_4(p)$

o : neighbor of q : $N_4(q)$

- ✓ m-adjacency is modification of 8-adjacency.
- ✓ It is used to remove the ambiguities of 8-adjacency.
- ✓ **This can be used to define single path between pixels.**

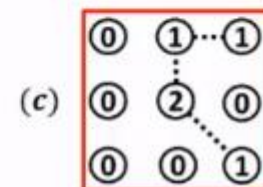
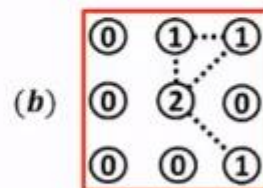
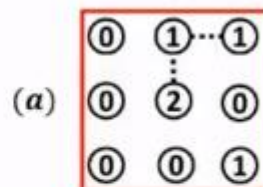
Connectivity

- V be the set of gray-level values used to define connectivity

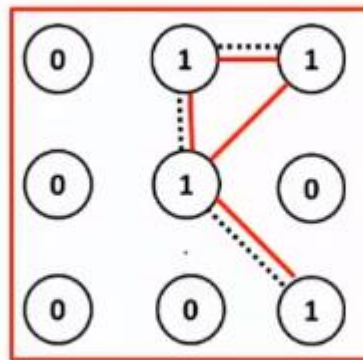
- Two pixels p, q that have values from the set V are:

- **4-connected**, if q is in the set $N_4(p)$
- **8-connected**, if q is in the set $N_8(p)$
- **m-connected**, iff
 - q is in $N_4(p)$ or
 - q is in $N_D(p)$ and the set $N_+(p) \cap N_+(q)$ is empty

$V = \{1, 2\}$



Adjacency / Connectivity



 8-adjacent

 m-adjacent

Paths and Path Lengths

- A path from pixel p with coordinates (x, y) to pixel q with coordinates (s, t) is a sequence of distinct pixels with coordinates:

$$(x_0, y_0), (x_1, y_1), (x_2, y_2) \dots (x_n, y_n),$$

where $(x_0, y_0) = (x, y)$ and $(x_n, y_n) = (s, t)$

$$(x_i, y_i) \text{ is adjacent to } (x_{i-1}, y_{i-1}), 1 \leq i \leq n$$

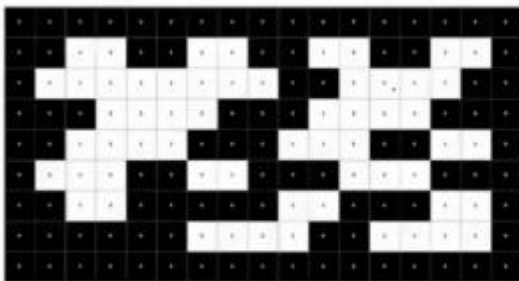
- Here n is the length of the path
- We can define 4-, 8-, and m-paths based on type of adjacency used

Connected Components.

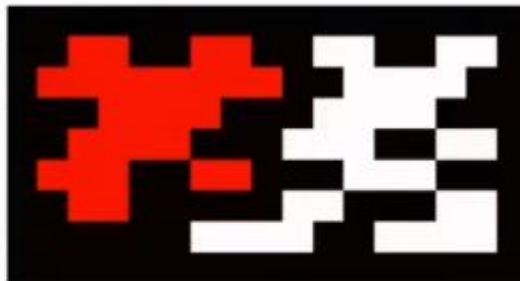
- **Connected pixels:**
 - If p and q are pixels of an image subset S then p is connected to q in S if there is a path from p to q consisting entirely of pixels in S
- **Connected Component**
 - For every pixel p in S , the set of pixels in S that are connected to p is called a connected component of S
- **Connected Set**
 - If S has only one connected component then S is called Connected Set

Connected Components - Examples

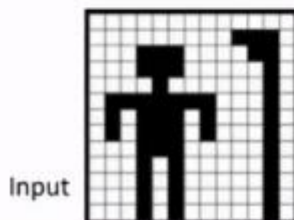
- **Connected-component labeling**



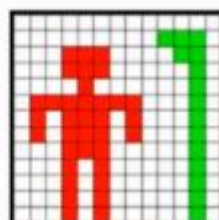
Input



Output: labeled image



Input



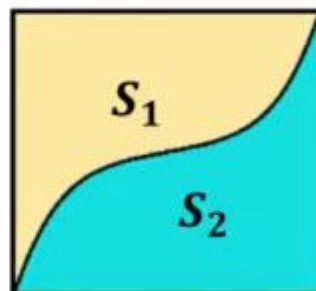
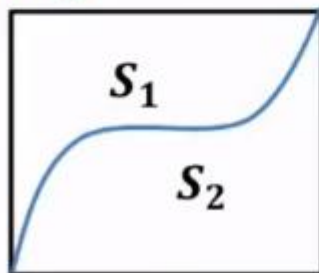
Output:
labeled image

Region and Boundaries

- **Region of the image**
 - A subset **R** of pixels in an image is called a *Region of the image* if R is a connected set
 - Each connected component forms a region
- **Boundary of the Region R**
 - It is the set of pixels in the region that have one or more neighbors that are not in R
- What if R happens to be entire Image?

Adjacency of Image Subsets

- Pixel p is adjacent to pixel q if they are connected (neighbor)
- Two image subsets S_1 and S_2 are adjacent if some pixel in S_1 is adjacent (neighbor) to some pixel in S_2



Arithmetic Logic Operation on Pixels

- Addition : $p + q$
- Subtraction: $p - q$
- Multiplication: $p * q$
- Division: p / q
- AND: $p \text{ AND } q$
- OR : $p \text{ OR } q$
- Complement: $\text{NOT}(q)$

Neighborhood based Operations on Pixels

- Value assigned to a pixel p_5 is a function of its neighboring pixels

	p_1	p_2	p_3	
	p_4	p_5	p_6	
	p_7	p_8	p_9	

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

$$p = w_1 p_1 + w_2 p_2 + w_3 p_3 + w_4 p_4 + w_5 p_5 + w_6 p_6 + w_7 p_7 + w_8 p_8 + w_9 p_9$$
$$= \sum_{i=1}^9 w_i p_i$$

Applications of Neighborhood Processing

- Tasks done using neighborhood processing:
 - *Smoothing / averaging*
 - *Noise removal / filtering*
 - *Edge detection*
 - *Contrast enhancement*

Issues in Neighborhood Processing

- **Choice of w_i 's**
 - N^2 values
- **Window Size**
 - Choice of N
- **Computation at boundaries**
 - *Do not compute at boundaries*
 - *Pad with zeros and extend image boundary*
 - *Pad assuming periodicity of image*
 - *Extrapolation of image*